

4. Liam and Simone are studying the distribution of oak trees in some woodland. They divided the woodland into 80 equal squares and recorded the number of oak trees in each square. The results are summarised in Table 1 below.

Number of oak trees in a square	0	1	2	3	4	5	6	7 or more
Frequency	1	4	21	23	13	11	7	0

Table 1

Liam believes that the oak trees were deliberately planted, with 6 oak trees per square and that a constant proportion p of the oak trees survived.

- (a) Suggest the model Liam should use to describe the number of oak trees per square. (2)

Liam decides to test whether or not his model is suitable and calculates the expected frequencies given in Table 2.

Number of oak trees in a square	0 or 1	2	3	4	5	6
Expected frequency	5.53	14.89	24.26	22.24	10.87	2.21

Table 2

- (b) Showing your working clearly, complete the test using a 5% level of significance. You should state your critical value and conclusion clearly. (7)

Simone believes that a Poisson distribution could be used to model the number of oak trees per square. She calculates the expected frequencies given in Table 3.

Number of oak trees in a square	0 or 1	2	3	4	5	6 or more
Expected frequency	12.69	16.07	s	14.58	t	9.37

Table 3

- (c) Find the value of s and the value of t , giving your answers to 2 decimal places. (4)

- (d) Write down hypotheses to test the suitability of Simone's model. (1)

The test statistic for this test is 8.749

- (e) Complete the test. Use a 5% level of significance and state your critical value and conclusion clearly. (3)

- (f) Using the results of these tests, explain whether the origin of this woodland is likely to be cultivated or wild. (2)



5. A factory produces pins.

An engineer selects 40 independent random samples of 6 pins produced at the factory and records the number of defective pins in each sample.

Number of defective pins	0	1	2	3	4	5	6
Observed frequency	19	11	7	2	0	1	0

(a) Show that the proportion of defective pins in the 40 samples is 0.15

(2)

The engineer suggests that the number of defective pins in a sample of 6 can be modelled using a binomial distribution. Using the information from the sample above, a test is to be carried out at the 10% significance level, to see whether the data are consistent with the engineer's suggested model.

The value of the test statistic for this test is 2.689

(b) Justifying the degrees of freedom used, carry out the test, at the 10% significance level, to see whether the data are consistent with the engineer's suggested model. State your hypotheses clearly.

(8)

The engineer later discovers that the previously recorded information was incorrect. The data should have been as follows.

Number of defective pins	0	1	2	3	4	5	6
Observed frequency	19	11	6	3	1	0	0

(c) Describe the effect this would have on the value of the test statistic that should be used for the hypothesis test. Give reasons for your answer.

(3)

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1. A researcher is investigating the number of female cubs present in litters of size 4. He believes that the number of female cubs in a litter can be modelled by $B(4, 0.5)$. He randomly selects 100 litters each of size 4 and records the number of female cubs. The results are recorded in the table below.

Number of female cubs	0	1	2	3	4
Observed number of litters	10	33	33	15	9

He calculated the expected frequencies as follows

Number of female cubs	0	1	2	3	4
Expected number of litters	6.25	r	s	r	6.25

- (a) Find the value of r and the value of s (3)
- (b) Carry out a suitable test, at the 5% level of significance, to determine whether or not the number of female cubs in a litter can be modelled by $B(4, 0.5)$. You should clearly state your hypotheses and the critical value used. (6)

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3. In a class experiment, each day for 170 days, a child is chosen at random and spins a large cardboard coin 5 times and the number of heads is recorded. The results are summarised in the following table.

Number of heads	0	1	2	3	4	5
Frequency	3	10	45	62	38	12

Marcus believes that a $B(5, 0.5)$ distribution can be used to model these data and he calculates expected frequencies, to 2 decimal places, as follows

Number of heads	0	1	2	3	4	5
Expected frequency	r	26.56	s	s	26.56	r

- (a) Find the value of r and the value of s (3)
- (b) Carry out a suitable test, at the 5% level of significance, to determine whether or not the $B(5, 0.5)$ distribution is a good model for these data. You should state clearly your hypotheses, the test statistic and the critical value used. (6)

Nima believes that a better model for these data would be $B(5, p)$

- (c) Find a suitable estimate for p (1)

To test her model, Nima uses this value of p , to calculate expected frequencies as follows

Number of heads	0	1	2	3	4	5
Expected frequency	2.07	14.65	41.44	58.63	41.47	11.74

The test statistic for Nima's test is 1.62 (to 3 significant figures)

- (d) State,
 (i) giving your reasons, the degrees of freedom
 (ii) the critical value
 that Nima should use for a test at the 5% significance level. (3)
- (e) With reference to Marcus' and Nima's test results, comment on
 (i) the probability of the coin landing on heads,
 (ii) the independence of the spins of the coin.
 Give reasons for your answers. (2)



3. Bags of £1 coins are paid into a bank. Each bag contains 20 coins.

The bank manager believes that 5% of the £1 coins paid into the bank are fakes. He decides to use the distribution $X \sim B(20, 0.05)$ to model the random variable X , the number of fake £1 coins in each bag.

The bank manager checks a random sample of 150 bags of £1 coins and records the number of fake coins found in each bag. His results are summarised in Table 1. He then calculates some of the expected frequencies, correct to 1 decimal place.

Number of fake coins in each bag	0	1	2	3	4 or more
Observed frequency	43	62	26	13	6
Expected frequency	53.8	56.6		8.9	

Table 1

- (a) Carry out a hypothesis test, at the 5% significance level, to see if the data supports the bank manager's statistical model. State your hypotheses clearly.

(10)

The assistant manager thinks that a binomial distribution is a good model but suggests that the proportion of fake coins is higher than 5%. She calculates the actual proportion of fake coins in the sample and uses this value to carry out a new hypothesis test on the data. Her expected frequencies are shown in Table 2.

Number of fake coins in each bag	0	1	2	3	4 or more
Observed frequency	43	62	26	13	6
Expected frequency	44.5	55.7	33.2	12.5	4.1

Table 2

- (b) Explain why there are 2 degrees of freedom in this case.
- (c) Given that she obtains a χ^2 test statistic of 2.67, test the assistant manager's hypothesis that the binomial distribution is a good model for the number of fake coins in each bag. Use a 5% level of significance and state your hypotheses clearly.

(2)

(2)